

**Undergraduate students have low fertility knowledge and high anxiety regarding future
fertility: an opportunity for education**

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24 **ABSTRACT**

25 **RESEARCH QUESTION:**

26 What are the childbearing intentions of undergraduate students and are they concerned about
27 their future fertility? Additionally, what do students understand about the age-related fertility
28 decline and are they interested in receiving fertility education?

30 **DESIGN:**

31 A cross-sectional, web-based survey that included 42 reproductive and fertility-related questions
32 was administered to actively enrolled undergraduates at Northwestern University.

34 **RESULTS:**

35 The survey was completed by 291 students with a mean age of 20.2 years. Of all participants,
36 62.5% plan to have children and 68.3% intend to delay childbearing. Significantly more females
37 than males (43.5% vs 4.5%, $P<0.001$) and premedical compared to non-premedical students
38 (50.4% vs 31.5%, $P=0.002$) reported having anxiety about future fertility due to career
39 aspirations. When surveyed about fertility knowledge, 25.4% stated that female fertility does not
40 dramatically decline until age 40 or later. When asked to estimate the oldest age a woman has
41 conceived using autologous oocytes, 83.3% reported age 48 or older. Of all participants, 72.3%
42 were interested in learning about fertility.

44 **CONCLUSIONS:**

45 The majority of surveyed undergraduates plan to delay childbearing, yet they have concerns
46 about how career goals will impact future fertility. Notably, females and premedical students

reported higher rates of anxiety when compared to their male and non-premedical counterparts. Knowledge about the age-related decline in fertility was limited, but students have a strong interest in learning about fertility, highlighting an opportunity for educational intervention at the undergraduate level. With education, students may be empowered to make informed decisions about future reproductive strategy earlier in time, potentially decreasing future anxiety.

KEY WORDS:

Fertility knowledge, anxiety, undergraduate students, education

INTRODUCTION

The mean maternal age at first birth in the United States has increased steadily due to multiple factors (Guzzo and Payne 2018, Londero *et al.* 2019, Martin *et al.* 2021). One major influence on this trend towards delayed childbearing is the pursuit of higher education (Mills *et al.* 2011, Nitsche and Bruckner 2021). Educational attainment is directly associated with age at first birth (Mills *et al.* 2011) and inversely associated with family size (Martinez *et al.* 2018). Well-educated individuals may be afforded career opportunities that alter reproductive decision-making, thereby leading to delays in family-building (Livingston 2015a). Although pursuit of higher education may be associated with delayed childbearing, fertility knowledge generally appears limited in well-educated individuals (Nouri *et al.* 2014, Danis *et al.* 2017, Bernardi *et al.* 2020). Undergraduates do not seem to fully comprehend the impact of female age on fertility as students have been shown to have overly optimistic expectations of reproductive success (Tyden *et al.* 2006). Additionally, despite undergraduates perceiving themselves as being educated about fertility, there is a notable lack of fertility awareness amongst university students (Peterson *et al.* 2012). If fertility awareness is lacking, and there is not an accurate understanding of how age impacts fertility, even well-educated individuals with an undergraduate degree may make reproductive decisions that will irreversibly alter their ability to successfully build their families. Thus, education on fertility is a critical intervention that may prevent future age-related infertility, particularly in those where educational and career pursuits may lead to delayed childbearing.

Education has proven to be beneficial in improving fertility knowledge (Daniluk and Koert 2015, Maeda *et al.* 2018). However, it is unclear whether undergraduate students are interested

receiving education on fertility and family building. The most desirable educational method to strengthen fertility knowledge in this population is also unknown. Moreover, it is uncertain whether students are concerned about their future fertility. Therefore, we developed this survey to assess undergraduate students' childbearing plans and concerns about future fertility, and to gauge interest in fertility education.

MATERIALS AND METHODS

Survey Development:

An online survey with 42 reproductive and fertility-related questions was created by the research team which included two full-time medical school faculty members who are board-certified Reproductive Endocrinologists. Research Electronic Data Capture (REDCap) was used to administer the survey. Participants were asked questions that ascertained their childbearing plans, personal concerns about fertility, baseline fertility knowledge, and interest in receiving fertility education. Thirteen of the fertility knowledge questions were taken from the Cardiff Fertility Knowledge Scale, which is a validated tool to evaluate fertility knowledge (Bunting et al. 2013). All survey questions can be found in the Supplemental Appendix. The Institutional Review Board at Northwestern University reviewed this study and deemed the study exempt (STU00214273). The study complies with all regulations and was conducted according to established ethical guidelines.

Participants

The survey was distributed through online communication tools and social media to students at Northwestern University. Only currently enrolled English-speaking undergraduates who were 18

years or older and residing in the United States were eligible to participate. Informed consent was obtained from all participants prior to beginning the survey. Participants who completed the survey were entered into a drawing to win one of five \$50 gift cards.

Statistical Analysis

REDCap was used to collect survey results and perform descriptive statistical analysis. Proportions are presented for categorical variables. Means with standard deviations are presented for continuous variables. All participants' responses were analyzed. Separate analyses were also completed to compare responses in males versus females, as well as premedical versus non-premedical students. Chi-square tests were performed to compare responses between groups. A *P*-value of < 0.05 was deemed statistically significant.

RESULTS:

Demographics

The survey was completed by 291 students (263 females and 28 males) with a mean age of 20.2 years. Other demographics are shown in Table 1. About half of respondents reported they were on a pre-professional track, and nearly 90% of those students reported being premedical.

Childbearing Plans

None of the participants had children, but 62.5% reported a desire to have children in the future. Of all participants, 68.3% reported plans to delay childbearing due to educational or career goals, yet almost 90% of this group intend to have their first child by age 30 (Figure 1). No participants reported plans to have their first child over the age of 35.

When males and females were compared in terms of their wishes for future children, there was no significant difference between groups as 68.2% of males and 62.2% of females reported a desire to have children. However, when asked about timing of childbearing, significantly more females reported being likely to delay childbearing due to educational or career aspirations (70.7% vs 40.9%, $P=0.004$). When survey responses were compared in premedical vs non-premedical students, significantly more premedical students reported wanting children (71.2% vs 55.2%, $P=0.008$). Furthermore, significantly more premedical students plan to delay childbearing due to educational or career aspirations (78.2% vs 60.1%, $P=0.002$).

Future Fertility Concerns

Of all participants, 46.9% reported thinking about their current fertility, 43.1% were worried about future fertility and 40.1% were anxious about future fertility due to career goals. Only 13.9% had anxiety about their partner status and the impact on fertility.

When the answers regarding fertility concerns were compared in males and females there were no differences in the proportion of students who worry about future fertility or had anxiety about fertility due to partner status. However, significantly more females reported thinking about fertility and having anxiety about future fertility due to career aspirations (Table 2). When responses between premedical and non-premedical students were compared, significantly more premedical students reported having anxiety about fertility due to career aspirations. There were no differences in the proportion of students who think or worry about future fertility, or who have anxiety about fertility due to partner status.

149 **Fertility Knowledge**

150 When knowledge related to age and fertility was assessed, 31.1% of participants reported that
 151 females are as fertile in their forties as they are in their thirties, 25.4% stated that female fertility
 152 does not dramatically decline until age 40 or later, 20.5% underestimated the risk of miscarriage
 153 in females 41 and older, and 21.5% reported that the IVF success rate is 80% in females under
 154 age 35. When asked to estimate the oldest age a woman has conceived using autologous oocytes,
 155 the mean age reported was 56 years and over 80% reported age 48 or older, as shown in Figure
 156 2. Finally, 37.4% of participants believed that a male's age never impacts reproductive
 157 outcomes.

159 **Interest in Fertility Education**

160 Of all participants surveyed, 72.3% reported they were interested in learning about fertility.
 161 When queried regarding their preferred method of receiving fertility education, the most popular
 162 choice was through educational videos (52.9%). In addition, 42.7% of students stated that they
 163 would like to learn about fertility as part of an undergraduate course. Of those who answered that
 164 they weren't interested in learning about fertility, 87.1% said that it was not important at this
 165 phase of life and 35.5% responded that they were not certain if they want to have children in the
 166 future.

168 **DISCUSSION:**

169 A majority of undergraduate students in this study plan to delay childbearing, yet they
 170 demonstrated limited knowledge regarding age-related decline in fertility and they over-
 171 estimated treatment success. Furthermore, our findings demonstrated that students, particularly

females and those who plan to enter a career in medicine, have anxiety about their future fertility due to their career goals. Finally, many undergraduate students are interested in receiving fertility education. This strong discordance between career aspirations and family building, combined with a reported desire for fertility education, highlights an important opportunity for primary prevention of age-related infertility through educational intervention at the undergraduate level.

The finding that the majority of respondents plan to delay childbearing due to career goals aligns with previous studies which revealed that individuals with a college degree had their first child at age ≥ 30 at significantly higher rates than those with less education.(Tyden *et al.* 2006, Martinez *et al.* 2018) However, in our study, nearly 90% of participants who plan to delay childbearing intend to have children by age 30. This perception contradicts real world data which shows only 44% of men and 41% of women with a bachelor's degree are likely to have children after the age of 30 (Livingston 2015a, Livingston 2015b). While most students acknowledged that pursuing a career or graduate education may delay childbearing, the vast majority believed a woman could conceive with her own oocytes 15 years beyond what is possible with IVF. Our results are corroborated by Peterson *et. al.* who found that undergraduates who reported they were well-educated also consistently overestimated when fertility declines(Peterson *et al.* 2012). This gap in fertility knowledge, and false sense of IVF success, may be contributing factors to future age-related infertility.

Despite plans to delay childbearing, our cohort demonstrated an overly optimistic and limited understanding of several crucial details related to fertility and reproductive biology. This issue is

pervasive and is in concert with prior studies in university students worldwide(Svanberg *et al.* 2006, Rovei *et al.* 2010, Hashiloni-Dolev *et al.* 2011, Virtala *et al.* 2011, Peterson *et al.* 2012, Mogilevkina *et al.* 2016, Sørensen *et al.* 2016). We found that 80% of participants incorrectly believed that the oldest maternal age of conception with an autologous oocyte was >48 years old. While the media often shows women conceiving via IVF at older ages(Sharma and Aggarwal 2016), the lack of transparency about the use of donor oocytes for conception is a disservice and provides false expectations (Feinberg 2017). This harkens to incomprehension of worsening oocyte quality with age and a misunderstanding regarding the necessity for oocyte donation to achieve successful pregnancy, especially after age 45.(Pokulniewicz *et al.* 2015) Our study also highlights that students overestimate IVF cycle success rates, which is concordant with previously published literature(Yu *et al.* 2016, Bernardi *et al.* 2020).

Multiple studies have revealed increased anxiety within the reproductive aged population, although none have specifically examined undergraduate students concerns about future fertility (Tyden *et al.* 2006, Daumler *et al.* 2016, Maeda *et al.* 2018, Na Nakhon *et al.* 2018). Our study demonstrated that a large proportion of participants experience worry, concern, and anxiety about fertility due to future career goals. Moreover, compared to their male colleagues, females are more concerned about their fertility and reported higher levels of anxiety. While prior studies support similar fertility knowledge gaps between sexes, and lower desire for parenthood in young adult males, this study is the first to show discordance in anxiety due to future fertility between undergraduate females versus males(Peterson *et al.* 2012, Mogilevkina *et al.* 2016, Sørensen *et al.* 2016). Furthermore, our study found that 20% more premedical students experience anxiety about fertility compared to students planning to pursue other careers.

Fertility Concerns and Education in Undergraduates

Specifically, being female and premedical resulted in the highest levels of anxiety related to future fertility. These findings are salient as peak fertility coincides with years spent in residency and fellowship training. Studies have demonstrated that approximately 25% of female physicians are diagnosed with infertility and are typically over the age of 30 at the time of first birth(Stentz *et al.* 2016). Notably, Stentz et al. revealed that 65% of female physicians endorsed their career had influenced childbearing decisions with approximately 30% desiring earlier conception and 17% indicating they would choose another specialty due to the impact on parenthood(Stentz *et al.* 2016). Given the 20-fold increase in women obtaining professional degrees over the past 40 years (Hickman *et al.* 2018), it is prudent to recognize that a large number of females may have future fertility concerns and that earlier education could decrease fertility-related anxiety and increase reproductive success.

This study highlights an opportunity to bridge the gap through undergraduate fertility education, with a focus on reproductive aging and the realistic success of fertility treatments. Prior studies have supported that fertility knowledge can be improved through a single educational intervention (Peterson *et al.* 2012, Wojcieszek and Thompson 2013, García *et al.* 2016). While promising, there is concern regarding the longevity of information retention and whether one educational intervention can have a meaningful impact for future reproductive decisions. While the best education modality warrants further investigation, it is likely that a longitudinal and multimodal approach, such as an undergraduate course, may be most effective for information retention. Given there are a paucity of studies on fertility educational intervention, research to examine various modalities in the university setting would be beneficial to determine optimal methods of disseminating information to this group of individuals.

Fertility Concerns and Education in Undergraduates

Our study is one of the first to identify concerns about future fertility and explore interest in fertility education (Svanberg *et al.* 2006, Rovei *et al.* 2010, Hashiloni-Dolev *et al.* 2011, Virtala *et al.* 2011, Peterson *et al.* 2012, Mogilevkina *et al.* 2016, Sørensen *et al.* 2016). A strength of this study is that the undergraduates surveyed were enrolled at a highly competitive institution comprised of a diverse student population from various geographic locations, ethnic backgrounds, and socioeconomic classes. Upwards of 20% of students pursue graduate/professional school or fellowships (Post Undergraduate Survey Data for (Northwestern Career Advancement 2020) making this cohort an ideal representation of previously described groups that delay childbearing (Mills *et al.* 2011).

One of the most important limitations of this study was the small sample size and inclusion of a single university that is not representative of all undergraduate students in the United States. Furthermore, a large percentage of participants plan to attend medical school, and these individuals may have had more fertility knowledge than other undergraduates, thus leading to an underestimation of the reported knowledge gap. Moreover, it is possible that the students who elected to participate in the survey were more concerned with their future fertility than the average student.

In summary, our data underscores fertility knowledge gaps in undergraduates and confirms that students have a poor understanding of reproductive aging. Age-related decline in fertility is an imperative biologic principle to comprehend, especially when delayed childbearing is planned in those seeking graduate degrees and post-doctoral fellowships. Fertility education at the

undergraduate level is warranted to improve reproductive success and empower decision making that can decrease anxiety related to future fertility.

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377 **Table Legend**

378 **Table 1.** Participant Characteristics

379 **Table 2.** Responses Regarding Fertility Concerns by Gender and Pre-Professional Track.

380

381 **Figure Legend**

382 **Figure 1.** Participants' estimation of the age at which they will have their first child.

383 **Figure 2.** Participants' estimation of the oldest age a woman has conceived using autologous

384 oocytes.

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